

Short Report: Sphere / Grassmann Distance Simulation

What was done

I simulated Gaussian factor models with latent dimension

$$k \in \{1, 2, 3\}, \quad n \in \{63, 126\}, \quad p \in \{100, 500, 1000, 2000, 5000, 10000\},$$

and fixed target radii

$$r \in \{0.1, 0.3, 0.5, 0.7, 0.9\}.$$

For each combination of (k, n, r, p) , I repeatedly generated sample data, estimated the top k sample eigenvectors, and recorded:

- sample-truth distance,
- sample-target distance,
- truth-target distance.

For $k = 1$, the target lies on the high-dimensional unit sphere and the distance is the spherical geodesic distance. For $k = 2$ and $k = 3$, the target is constructed on the Stiefel manifold through principal-angle coordinates and the distance is the Grassmann geodesic distance.

Factor model parameters

The noise level is $\sigma = 1$. The factor-model eigenvalues used in the code are:

- 1-factor: (9.0)
- 2-factor: (9.0, 4.0)
- 3-factor: (9.0, 4.0, 2.0)

The data matrix is generated as

$$X = U \operatorname{diag}\left(\sqrt{\lambda_1 - \sigma^2}, \dots, \sqrt{\lambda_k - \sigma^2}\right) F + \sigma E.$$

Results

Figure 1 shows the 1-factor experiment with spherical geodesic distance. Each panel corresponds to one radius and one sample size.

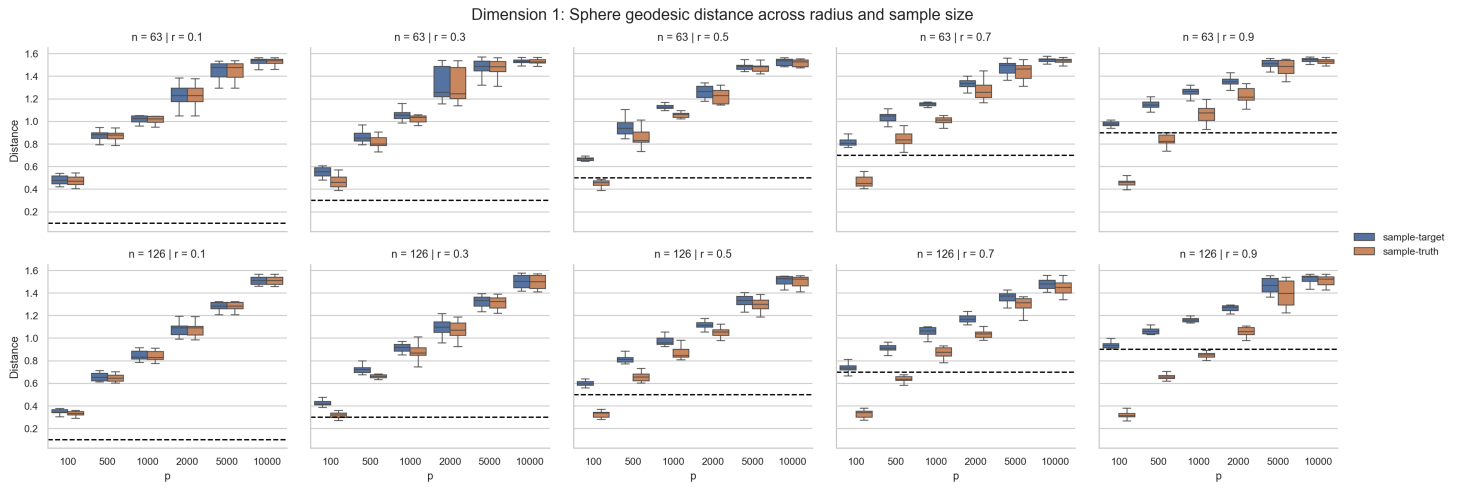


Figure 2 shows the 2-factor experiment with Grassmann distance on the Stiefel/Grassmann setting. Each panel corresponds to one radius and one sample size.

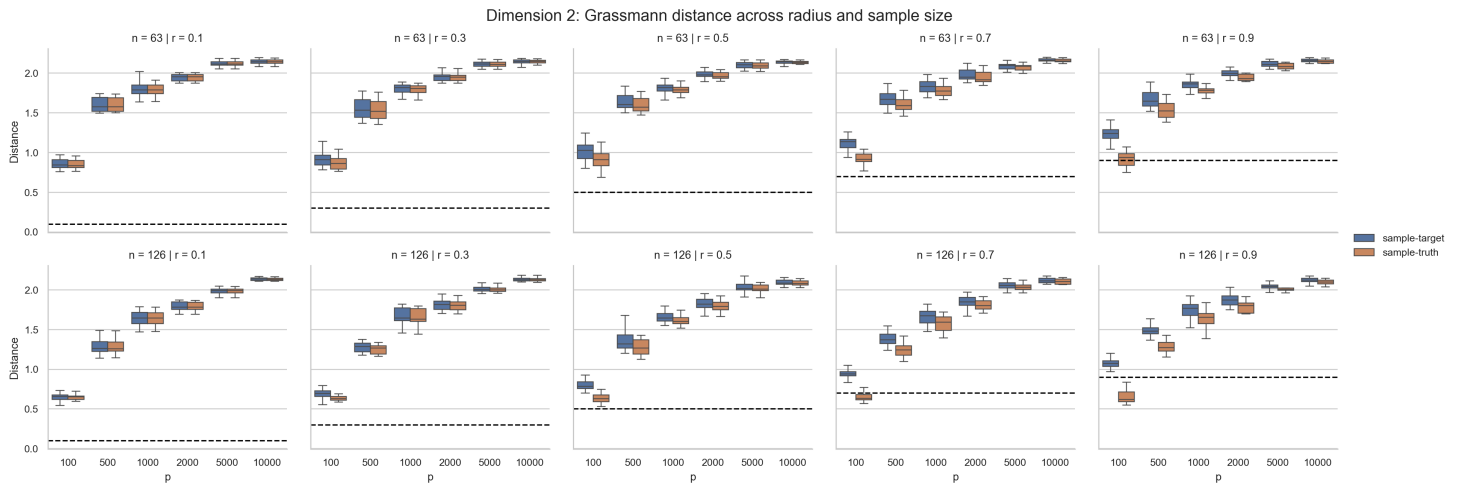


Figure 3 shows the 3-factor experiment with Grassmann distance on the Stiefel/Grassmann setting. Each panel corresponds to one radius and one sample size.

Dimension 3: Grassmann distance across radius and sample size

